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Source depth estimation based on synthetic aperture beamforming for a moving source

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A continuous wave signal received on a single hydrophone from a moving source is beamformed using the synthetic aperture created by the source, where the signal at each range is steered by a range-dependent phase, relative to the starting point. The range increment (aperture) is determined based on the Doppler shift estimated from the data, knowing the original signal frequency. Given a sufficient signal-to-noise ratio, the source depth can be estimated from the beam output, assuming knowledge of the mode depth functions based on the nominal sound speed and bottom profile in the area. The method is illustrated with simulated data and at-sea data. For real data, the signal phase contains a random, incoherent component caused by the (random) source motion and media fluctuations in addition to the deterministic range-dependent component due to source range change. A phase locked loop is introduced to remove the random component assuming that the random component fluctuates faster with time than the range-dependent phase. When a vertical array of receivers are available covering the depth span of interest, the beam output can be used directly to estimate the source depth. In this case, no knowledge of the acoustic environment is needed.

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I. INTRODUCTION

Acoustic methods for localizing a target, particularly a submerged target, in an underwater acoustic waveguide are a topic of great interest. A popular method for estimating the source range and depth is the matched field processing (MFP)^{1,2} or matched mode processing (MMP)^{3,4} (also known as modal beamforming), which require a vertical line array (VLA), or an array with an effective vertical aperture [such as a horizontal line array (HLA) in the endfire direction⁵], and also knowledge about the environment [such as the sound speed profile (SSP) and bottom profile]. Source localization generally requires simultaneous searches for the source range and depth using the range-depth ambiguity function. One often finds an erroneous estimate of the source depth given an incorrectly estimated source range and vice versa, as is the case when high sidelobes are present. The sidelobe problem is greatly exemplified when the assumed environment differs from the true environment, referred to as the environmental mismatch problem. Estimation of the source depth or range is in general not robust, when the estimation of the one depends on the other.

For some applications, such as determining if a potential target is submerged or not, an approximate estimation of the source depth will suffice, with or without knowledge of the source range. (In passive listening, the source range is often not known and difficult to estimate.) This is known as the depth classification or discrimination problem. This paper proposes a method to estimate the source depth of a moving source, traveling at a constant depth and speed in a range-independent environment, without knowing the source

range. (The generalization to other cases is outside the scope of this paper.) It assumes a continuous wave (CW) signal with a high signal-to-noise ratio and is intended for low frequency signals where the effect of the medium fluctuation is less significant. The method applies array beamforming to the moving source data, based on the synthetic aperture created by the source motion. The signals at various ranges are multiplied by a range-dependent steering vector; absolute range is not known. The beam output is used to estimate the source depth.

The application of synthetic beamforming to estimate the source depth based on data received on a single receiver is new, to be the best of the author's knowledge. As remarked above, in order to estimate the source depth using the MFP or MMP, one needs a VLA, and must know or search for the source range. In order to estimate the source range using a normal mode propagation model, one must know the mode wavenumbers to a high precision (e.g., the third or fourth decimal point). The mode wavenumbers are sensitive to the sound speed changes, and hence MFP/MMP requires precise and timely knowledge of the SSP and (prior) knowledge of the ocean bottom, else the results are often unreliable (the environmental mismatch problem). The proposed method does not require knowledge of, nor needs to estimate the source range, hence it is thus more robust than the MFP or MMP method. It does require knowledge of the mode depth functions. For approximate depth estimation, the mode depth functions based on the nominal sound speed and bottom profile will often suffice. The proposed method, developed for a single receiver, can be extended to a VLA (Sec. IV B). In this case, the source depth is estimated directly from the beam output as a function of depth. No knowledge of the environment is needed.

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II. SYNTHETIC APERTURE MODAL BEAMFORMING

We shall consider a range-independent environment and assume that the source moves away from the receiver at a constant speed during the observation time. The source range is unknown and its depth can be slowly changing with time; for the present discussion we shall assume it is at a fixed depth. We assume that the source radiates a tonal signal, and the source speed can be estimated based on the Doppler shifts of the signal, $v = C(\Delta f/f)$, where C is the average sound speed, Δf is the Doppler shift, and f is the known source frequency. The received data as a function of time are implicitly related to the source range by $r = r_1 + vt$, where r_1 is the unknown source range at $t = 0$. During the observation time T , the source travels a distance $\Delta R = vT$, which is known and is the aperture of the synthetic array. The pressure field at different ranges is beamformed as follows:

$$g(k_r, z_r) = \frac{e^{i\pi/4}}{\sqrt{2\pi k_r}} \int_{r_1}^{r_2} p(r, z_r) e^{ik_r r} S(r) dr, \quad k_r r_1 \gg 1, \quad (1)$$

where $p(r, z_r)$ is the signal received at a receiver at a depth z_r and range $r = r_1 + vt$, where $t = t_n$, for the n th block of data, $e^{ik_r r}$ is the steering vector, and $S(r)$ is the weighting coefficient for the data (known as the array shading coefficient for plane wave beamforming). The function $S(r)$ is intended to compensate for the spreading loss and will be determined from data using

$$S(r) = \langle |p(r, z_r)|^2 \rangle^{-1/2}, \quad (2)$$

where the angular brackets $\langle \rangle$ denote range averaging and/or smoothing operation. Note that if the source range is known, by setting $S(r) = \sqrt{r}$, Eq. (1) reduces to the Hankel transform.⁶ In this case, $g(k_r, z_r)$ is known as the depth-dependent Green's function. Range averaging/smoothing is required in the present case to obtain an approximate $\sqrt{r}$ dependence as discussed in Sec. III. Errors induced by this approximation will be evaluated using simulated data.

To guide the discussions, let us express the pressure field in terms of the normal modes as follows:

$$p(r, z_r) = \sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r}} \phi_m(z_s) \phi_m(z_r) e^{-ik_m r - \alpha_m r - i\pi/4}, \quad (3)$$

where $\phi_m(z)$ is the mode depth function, k_m is the mode wavenumber and α_m is the mode attenuation coefficient. Assuming that $S(r)$ is approximately proportional to $\sqrt{r}$, one finds, after substituting the normal mode expression into Eq. (1), that

$$\begin{aligned} g(k_r, z_r) &\simeq \sum_{m=1}^M \frac{\phi_m(z_s) \phi_m(z_r)}{\sqrt{k_r k_m}} \int_{r_1}^{r_2} e^{i(k_r - k_m)r - \alpha_m r} dr \\ &= \sum_{m=1}^M a_m \frac{\phi_m(z_s) \phi_m(z_r)}{k_r - k_m - i\alpha_m}, \end{aligned} \quad (4)$$

where $a_m = [e^{i(k_r - k_m)r_2 - \alpha_m r_2} - e^{i(k_r - k_m)r_1 - \alpha_m r_1}] / i k_m$. The wavenumber spectrum $g(k_r, z_r)$ has many peaks at the modal wavenumbers.^{6,7} The spectral peak at $k_r = k_m$ is given by

$$g(k_m, z_r) = b_m \phi_m(z_r), \quad (5)$$

where $b_m = (2e^{-\alpha_m r_0}) / (\alpha_m k_m) \sinh[(\alpha_m \Delta R)/2] \phi_m(z_s)$ is a real number, with $r_0 \equiv (r_2 + r_1)/2$. Realizing that b_m contains information about the source depth z_s , we define a source-depth ambiguity function for a single receiver by

$$D(z) = \left| \sum_{m=1}^M \phi_m(z) \frac{g(k_m, z_r)}{\bar{\phi}_m(z_r)} \right|^2, \quad (6)$$

where $\bar{\phi}_m(z_r)$ is a regularized mode depth function given by $\bar{\phi}_m^{-1}(z_r) = [\phi_m(z_r)] / [\phi_m^2(z_r) + \Delta^2]$, where Δ is chosen empirically (on the order of one tenth of the maximum value of the mode depth functions) to prevent blow up of the inverse mode depth function at zero-crossing points. In this approach (given only a single receiver), the mode depth function $\phi_m(z)$ will be estimated using a nominal SSP; no knowledge of mode wavenumbers is required. Note that the mode depth functions are insensitive to the sound speed fluctuations over the observation time, and the ones estimated from the nominal SSP are generally good enough for coarse estimation of the source depth. [The depth estimation error is due to mismatch between the calculated mode depth functions $\phi_m(z_s)$ and the true mode source depth functions $b_m \sim \phi_m^{\text{data}}(z_s)$ at the source depth. The depth ambiguity functions $|\sum_{m=1}^M \phi_m(z) \phi_m(z_s)|^2$ and $|\sum_{m=1}^M \phi_m(z) \phi_m^{\text{data}}(z_s)|^2$ differ slightly in shape since $\phi_m(z_s)$ and $\phi_m^{\text{data}}(z_s)$ are real and insensitive to the sound speed variations. Consequently, one expects a "small or moderate" error in the source depth estimation. In practice, the depth resolution is limited by the hydrophone spacing (due to finite number of receivers used), which is often larger than the depth estimation error due to SSP uncertainty.] To compare the depth distributions for sources at different depths, we define a normalized depth distribution function, given by Eq. (6) divided by the sum of $D(z)$ over the depth grids or sample points used in the depth search/estimation.

The above method is illustrated using simulated data given in Ref. 7 for a downward refractive SSP. (The readers are referred to Ref. 7 for details.) We consider two sources, one at depth of 50 m, referred to as the deep source, and another at a depth of 4 m, referred to as the shallow source and one receiver. To study the performance as a function of the receiver depth, we consider two cases: the first case involves a receiver located at a depth of 70 m, and the second case considers a receiver located at a depth 18 m. Assume each source moves away from the receiver with a speed of 5 kn, transmitting a narrowband signal at 350 Hz. (The wavenumber estimation is governed by the synthetic aperture, not the speed.) We will calculate the depth ambiguity function based on data received from each source and determine whether one can discriminate the deep source against the shallow source. In the simulation, we assume that the source starts out at a range of 5010 m, but this information is unknown to the receiver, nor is it critical (see discussions in the following paragraph). The received pressure field is sampled every second.⁷ The pressure field data sampled over

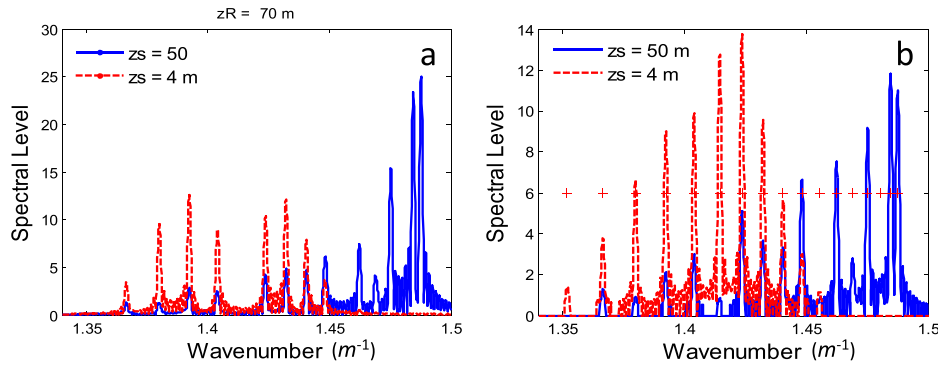


FIG. 1. (Color online) Modal wave-number spectra estimated using synthetic aperture beamforming based on signals from a shallow ($z_s = 4$ m) and deep ($z_s = 50$ m) source: (a) for a single receiver at 70 m depth, and (b) summed over the VLA. The “+” symbols denote the true modal wavenumbers.

time, covering a range span of 4990 m, are used to estimate the wavenumber spectrum for each receiver using Eq. (1). The wavenumber spectrum estimated using Eq. (1) for a receiver at 70 m depth is shown in Fig. 1(a) for each of the two sources; the symbols “+” denote the true mode wavenumbers used to identify the mode number for each spectral peak. Figure 1(a) shows that the deep source (solid lines) and shallow source (dashed lines) excite different groups of modes, providing the basis for depth classification. To estimate the source depth, we calculate the depth ambiguity functions, Eq. (6), based on the wavenumber spectrum deduced from a single receiver data. We assume knowledge of the mode depth functions, which are calculated using the given SSP (Fig. 1, Ref. 7). The normalized depth distribution functions for the deep and shallow sources are shown in Fig. 2(a) for the receiver at a depth $z_r = 70$ m. One finds that the depths of the two sources are correctly estimated based on the depth distribution functions. For the receiver at a depth $z_r = 18$ m, the normalized depth distribution functions are shown in Fig. 2(b). The distribution for the deep source has higher sidelobes, but the source depth is still reliably estimated.

How is the performance influenced by the unknown initial source range? One notes that sources at different ranges will excite modes somewhat differently since mode attenuations are range-dependent. For synthetic beamforming, the dependence on the initial source range is through $e^{-\alpha_m r_0}$ in b_m [see the derivation leading to Eq. (5)]. Consider, for example, two different data sets where the source travels the same range span but starts out at different initial ranges, r_1 and r'_1 . One finds that the spectral terms are related by $b_m(r'_1) = b_m(r_1)e^{\alpha_m(r'_1 - r_1)}$. In practice, $|r'_1 - r_1|$ is on the order of km or less (depending on the detection range). For the simulated data, α_m is on the order of 10^{-4} – 10^{-5} for the

low (<10) order modes. For these modes, one finds that $b_m(r'_1) \simeq b_m(r_1)$. For the high (10–20) order modes, b_m may change by $\sim 30\%$ for $|r'_1 - r_1| \sim 4$ km, but these modes are weakly excited and the effect on mode depth function is negligible.

III. SWellEx96 DATA ANALYSIS

The SWellEx96 experiment was conducted near San Diego, CA in the spring of 1996. The environmental information and data are available online.⁸ In this section, we analyze 35 min of data, 5 min after the start of event S5, in which the source-ship traveled along an isobath of a mildly sloping environment at a distance from 8 to 3 km from the receiving VLA.⁸ The water depth was approximately 210 m. The data analyzed involved two tonal signals, one at 127 Hz projected by a shallow source ($z_s = 9$ m), and the other at 130 Hz projected by a deep source ($z_s = 54$ m). The data were received simultaneously on a VLA. We shall focus on the signals received on the bottom most hydrophone on the VLA to see how well the proposed method can estimate the two source depths. We assume knowledge of the mode depth functions, which are calculated based on the SSP observed during this event.⁸

As remarked in Ref. 7, real data requires signal preconditioning as it is unrealistic to expect a ship to travel with a constant (radial) velocity over a long time. The motions by the surface waves are known to induce instantaneous acceleration and deceleration on the ship, causing jittering motion on the sources towed behind. Based on the GPS data, the range from the ship to the VLA is shown in Fig. 3(a) for the time period of interest. The data were interpolated (using the Matlab interpolation program) to calculate the velocity as shown in Fig. 3(b). While the estimated velocity may not be

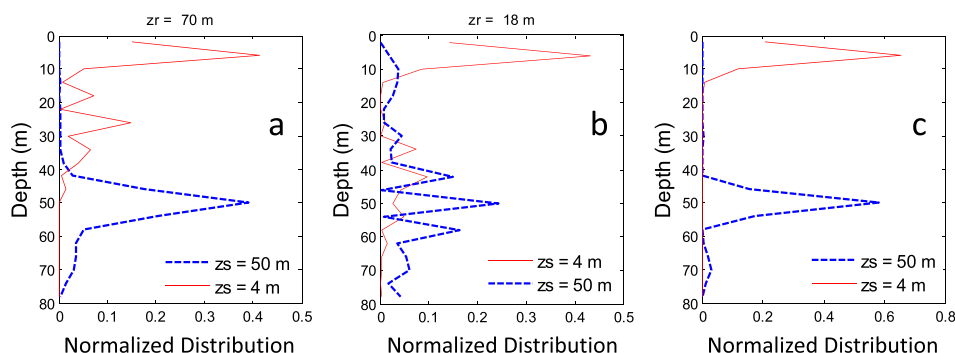


FIG. 2. (Color online) Normalized depth ambiguity functions for the shallow ($z_s = 4$ m) and deep ($z_s = 50$ m) sources using data collected: (a) from a single receiver at 70 m depth, (b) from a single receiver at 18 m depth, and (c) using a VLA.

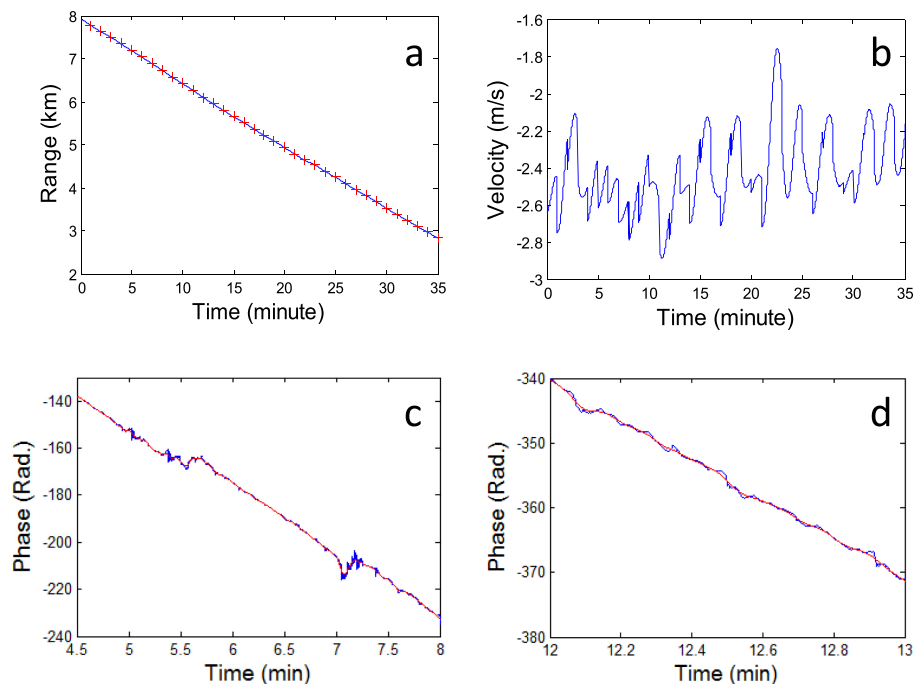


FIG. 3. (Color online) (a) Ship range to the VLA as a function of time. (b) Velocity of the ship deduced from (a). (c) and (d) Phase of the 127 Hz signal for two different time periods; wiggly lines show the original phase and smooth lines show the filtered phase.

accurate, it does indicate that the ship encounters instantaneous acceleration and deceleration.

For the source traveling with a constant (radial) velocity, the range change is known to induce a linearly time-varying signal phase, known as the Doppler shift. Random instantaneous acceleration and deceleration on the source introduce a (oscillating) time-varying phase on the projected signal, leading to Doppler broadening of the signal frequency.⁹ The random phase (for even a small amount of Doppler broadening) can severely degrade the ability to estimate/resolve the mode wavenumbers using the synthetic aperture beamforming method. The same data were processed by Walker *et al.*⁹ who addressed the effect of radial source accelerations on the estimation of modal frequency shifts. They found “Doppler broadening due to radial source accelerations degrades the effectiveness of the (mode) extraction method,” requiring a technique to compensate for Doppler broadening in order to recover the original signal phase for a source moving with a constant (radial) velocity.

The fluctuating ocean media also introduce a random phase on the received signal. The random media effects on sound propagation are expected to be not so significant at low frequencies. (MFP assumes that the signal phases between the receivers can be modeled as a function of depth using the so called replica field.) We assume that the phase distortion due to jittering source motion and the random media vary relatively fast with time, faster than the phase change due to source range change, which varies slowly with time for a source traveling at a low speed. To remove the random (oscillating) phase, we propose: (1) a second order phase locked loop (PLL)¹⁰ to estimate the time varying phase of the tonal signal $\tilde{\theta}(t)$, defined by $p(t) = Pe^{-i2\pi f_c t + i\tilde{\theta}(t)}$, where P is treated as a constant, and (2) a low pass filter to remove the oscillating/jittering phase, yielding $\bar{\theta}(t)$, and an estimate of the random phase $\hat{\theta}(t) = \tilde{\theta}(t) - \bar{\theta}(t)$. The PLL has been widely used

in radio, telecommunications, computers and other electronic applications. In earlier days, it was implemented with a variable frequency oscillator and a phase detector. The oscillator generates a periodic signal. The phase detector compares the phase of that signal with the phase of the input periodic signal and adjusts the oscillator to keep the phases as closely matched with the input signal as possible. Through the feedback loop it tracks the phase between the input and output signal. Today, the PLL is mostly implemented digitally. In this work, we shall use a second order phase locked loop which employs a proportional tracking coefficient K_1 and an integral tracking coefficient K_2 as given below

$$\tilde{\theta}(n+1) = \tilde{\theta}(n) + K_1 \varphi(n) + K_2 \sum_{m=0}^n \varphi(m), \quad (7)$$

where $\varphi(m)$ is the phase error between the input signal (data) and the oscillator at time m , and $\tilde{\theta}(n)$ is the output phase. We use $K_1 = 0.001$ and $K_2 = 0.0001$ as commonly used in underwater acoustic communications.¹¹

The data are first processed by Fourier analysis to estimate the Doppler shift for each tonal signal. One finds an average Doppler shift of ~ 0.21 Hz, yielding a target speed of 2.42 m/s which is consistent with the estimated speed of 2.43 m/s based on GPS log. For each signal of interest, applying a band pass filter centered on the Doppler-shifted frequency, with a bandwidth of 1 Hz, we obtain a time series for the corresponding (narrowband) signal, upon which we next apply the phase locked loop and a low pass filter to remove the phase jitter. Figures 3(c) and 3(d) show the phase as a function of time over two different time scales for the 127 Hz signal—the phase time series measured using the PLL are shown by wiggly lines, and the filtered phase time series are shown by the smooth lines. The rapid phase change (with period < 0.2 min) is attributed to the source

motion and media fluctuation and is low pass filtered. The filtered phase is used to construct the filtered signal time series from moving source as if it were moving at a constant speed. The time series is divided into blocks, each with 4096 samples, with a time duration of ~ 2.73 s, corresponding to a range increment of $\Delta r \simeq 6.63$ m. The pressure field $p(r_n, z_r)$ is next estimated for each block of data [to be used as input to Eq. (1)] according to the procedure given in the Appendix, where $r_n = r_1 + n\Delta r$, with an unknown initial range r_1 . The Appendix also provides an error analysis. To estimate $S(r)$ in Eq. (1), the signal intensity $|p(r_n, z_r)|^2$ was range averaged to produce a smooth curve. It was then fitted in the log domain by a linear function. The resulting $S(r)$ is used in Eq. (1) to compensate for the spreading loss. The above processing is applied to the 127 and 130 Hz signals separately. The resulting wavenumber spectra using Eq. (1) and the depth distribution functions using Eq. (6) will be presented below.

To guide the data analysis, we first create simulated data for event S5 assuming a source run with a range span of ~ 5 km using the SWellEx96 environment as posted on the website.⁸ Figure 4(a) shows the mode wavenumber spectra deduced from the simulated data received on a sensor at the ocean bottom for the 127 Hz signal projected by the shallow ($z_s = 9$ m) source and the 130 Hz signal projected by the deep ($z_s = 54$ m) source. Figure 4(b) shows the normalized depth distribution function based on the single-receiver algorithm, Eq. (6), for both signals. As expected, the depth ambiguity function for the 127 Hz signal shows a peak at a depth of ~ 10 m and the depth ambiguity function for the 130 Hz signal shows a peak at ~ 56 m in agreement with the true source depth (within the depth resolution). Next we present the result using the SWellEx96 data. Figure 4(c) shows the mode wavenumber spectra estimated from the bottom-most receiver for both signals using Eq. (1). Figure 4(d) shows the

corresponding normalized depth distribution functions using Eq. (6) for both signals. One finds that the depth ambiguity function at 127 Hz peaks at 8 m and the depth ambiguity function at 130 Hz peaks at 38 m. The source depths of the two signals seem to be separately estimated. Note that to calculate the depth ambiguity function using Eq. (6), one needs to identify the mode number associated with the spectral peaks—some spectral peaks are treated as sidelobes and not selected. We use the calculated modal wavenumber spectrum (based on simulated data) as a guide to help with the mode identification. We note that the mode wavenumber decreases monotonically with the mode number. The absolute value of the difference of the mode wavenumbers between adjacent modes increases monotonically with the mode number for the low order (e.g., 1–20) modes. Although the mode wavenumbers are not precisely known due to uncertainty in the SSP and bottom, the mode wavenumber differences (between the adjacent modes) and the rate of increase are approximately known (to the first valid digit). [For the present case, the mode wavenumber difference increases approximately linearly with the mode number for the low order (< 20) modes.] Now, if the first mode is excited (as for a submerged target), since the first mode has the largest wavenumber, the right most spectral peak in the data will be associated with the first mode. As an example, compare the solid line in Fig. 4(c) with that in Fig. 4(a). Once the first mode is identified, the rest follows naturally (by matching the data with the simulated data by the rate of increase of the mode wavenumber differences). If only higher order modes are excited (as for a surface ship), one can identify the modes by estimating the wavenumber spacing, and matching that with the calculated mode wavenumber spacing. This mode identification process involves (admittedly) an educated guess since the wavenumber spacing (and its rate of increase) is not precisely known. (For a

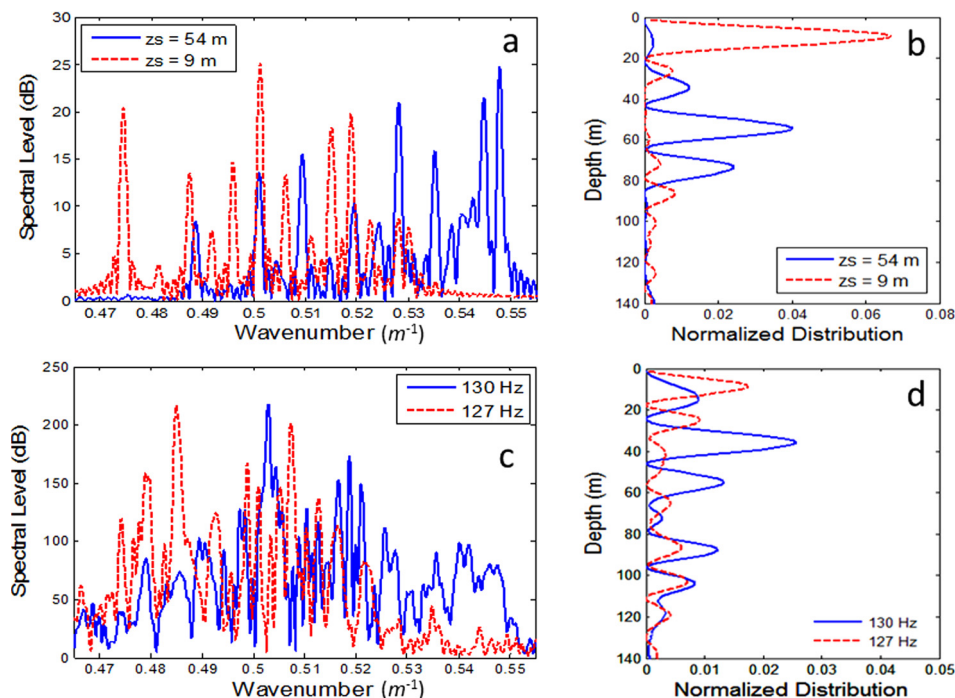


FIG. 4. (Color online) Modal wavenumber spectra for (a) simulated data and (c) at-sea data. Normalized depth ambiguity functions estimated using (b) simulated data and (d) at-sea data. The shallow source ($z_s = 9$ m) transmitted a tonal signal at 127 Hz and the deep ($z_s = 54$ m) source transmitted a tonal signal at 130 Hz.

crude estimate of the depth error, we note that mode identification error between the n and m modes is expected to produce a depth estimation error on the order of $H/n-H/m$, where H is the water depth. Which mode is in error is case-dependent, so is the sensitivity analysis.) Also, as discussed in the Appendix, one finds that the measured wavenumbers are subject to the uncertainty in the Doppler shift estimation. In the data analysis, we identify the modes by fine-tuning the residue Doppler shift (or shifting the measured wavenumber spectrum up or down the wavenumber axis) until one achieves a good match (correlation) between the data spectrum and calculated wavenumber spectrum. In practice, one can use the signals from either an augmented source or a known surface ship as training data to help the mode identification.

Comparing the depth ambiguity function estimated from data [Fig. 4(d)] with that estimated from the model [Fig. 4(b)], one finds the peak at 8 m in the data is smaller than that shown in the simulation. The reason is because there is an aliased peak close to the ocean bottom in the depth ambiguity function estimated from the data (not shown here). It seems that a near-surface and near-ocean-bottom source excites similar groups of modes which are resolved in simulated data but not so in the at-sea data. In that case, one will have a “grating” lobe. In practice, this is not likely to present a problem as no moving target is expected near the ocean bottom. For the deep source, one notices that the depth ambiguity function has only one peak; the source is “uniquely” localized. Comparing the performance, one finds that the main lobe level in the data [Fig. 4(d)] is somewhat lower than that found in the simulation [Fig. 4(b)]. This is because the real data have more sidelobes [Fig. 4(d)]. Note the horizontal scales in Figs. 4(b) and 4(d) are slightly different.

IV. SYSTEM IMPLICATIONS

A. HLA

The above method assumes a strong signal. The method can in principle be applied to a weak signal given a HLA, whereby the signal is enhanced by array gain when steered to the target direction (bearing). This creates an ability for the HLA to distinguish a submerged target from a surface ship (even) in the broadside directions of the array. This capability did not exist before.

B. VLA

Given a VLA that spans the water column, one can apply synthetic beamforming to the moving source data for each receiver. In this case, one finds that the source depth can be estimated directly from data without any knowledge of the mode depth function nor the SSP. For a VLA of many receivers, the weighting coefficient in Eq. (1) will be estimated using

$$S(r) = \left\langle \left[\sum_{j=1}^N |p(r, z_j)|^2 \Delta z \right] \right\rangle^{-1/2}. \quad (8)$$

In this case, one finds $S(r) \sim \sqrt{r}$ ignoring small corrections due to differences in mode attenuation coefficients.⁷ The source-depth ambiguity function is given by the beam power for each receiver. Using Eq. (5), one finds

$$D(z_j) = \left| \sum_{m=1}^M g(k_m, z_j) \right|^2 = \left| \sum_{m=1}^M \phi_m(z_j) \phi_m(z_s) \delta_m \right|^2; \quad j = 1, 2, 3, \dots, N, \quad (9)$$

where z_j is the receiver depth of the j th hydrophone, and $\delta_m = (2e^{-\alpha_m r_0}) / (\alpha_m k_m \sinh[(\alpha_m \Delta R)/2])$ is referred to as the mode shading coefficient.¹³ Equation (9) shows that the depth ambiguity function is related in theory to the mode depth functions and can be used to estimate the source depth; it is related to the MFP/MMP depth function evaluated at the depths of the receivers as shown below in the subsection, Sec. IV B 1. Experimentally, the depth ambiguity will be obtained by performing the synthetic aperture beamforming analysis on the received data for each hydrophone. The spectral energy [the second term of Eq. (9)] is displayed as a function of the hydrophone depth. The depth where the depth ambiguity function is the largest is the estimated source depth. No mode depth functions are used in this analysis, hence no knowledge of the propagation environment is required. To estimate the probability distribution function of the source depth, we define $P(z|z_s) = KD(z)$, for $z = z_j$, $j = 1, 2, 3, \dots, N$, where $K = [\sum_{j=1}^N D(z_j)]^{-1}$ is a normalization factor. The depth resolution is limited by the receiver separation Δz .

The above method is illustrated using simulated data given in Ref. 7 (See Sec. II and Ref. 7 for details.) The VLA contains 20 element spaced at 4 m covering the water column from 2 to 80 m. The wavenumber spectrum, Eq. (1) summed over all receivers, is shown in Fig. 1(b) as a function of wavenumber. The normalized depth ambiguity function is shown in Fig. 2(c) for both the deep and shallow sources. Again, the depth ambiguity functions are deduced directly from the simulated data without assuming any knowledge of the SSP and/or the bottom profile. This method can also be applied to multiple receivers deployed at different depths, but not necessarily at the same location (such as a slanted VLA), since the data are independently processed. To estimate the source depth, the depth span of the receivers should cover the depth where the target is expected to travel, and be sufficiently wide to see a distribution. (The target depth should be within the search depths.) For SWellEx96 data, the receiving VLA is located at the lower half of the water column whereas the towed source is in the upper half of the water column. The model-independent depth estimation method, Eq. (9), cannot be applied here.

1. Relation with the MFP

In this section, Eq. (9) will be shown to be the similar to the matched-mode ambiguity function evaluated at the source range. Recall the matched-mode range-depth ambiguity function given by³ $DMMP(r, z) = (r/2\pi) \left| \sum_{m=1}^M [A_m^{rplc}(r, z)] * A_m^{\text{data}} \right|^2$,

where $A_m^{\text{data}} = \sum_{j=1}^N \phi_m(z_j) p(r_s, z_j) \Delta z$ is the m th mode amplitude estimated from the data, $p(r_s, z_j)$ is the pressure field on a receiver at depth z_j from a source at range r_s , Δz is the phone spacing, and $A_m^{\text{rplc}}(r, z) = \sqrt{(2\pi/rk_m)} \phi_m(z) e^{ik_m r}$ is the m th mode amplitude of the replica field for a hypothetical source located at (r, z) , where k_m is the complex wavenumber for the m th mode. Given a VLA spanning the water column, performing the mode decomposition using the mode orthogonality condition, one obtains $A_m^{\text{data}}(r, z) = \sqrt{(2\pi/rk_m)} \phi_m(z_s) e^{ik_m r_s}$. If the source range is correctly estimated, $r = r_s$, one finds

$$\text{DMMP}(r = r_s, z) = \left| \sum_{m=1}^M \phi_m(z) \phi_m(z_s) \delta'_m \right|^2, \quad (10)$$

where $\delta'_m \equiv e^{-2\alpha_m r_s} / k_m$. One notes that the depth ambiguity function of the proposed method, Eq. (9), is similar to the matched-mode ambiguity function evaluated at the source range, Eq. (10). The difference lies in the use of the mode shading coefficients, which modifies the shape of the distribution function and usually does not impact (the accuracy of) the depth estimation. (The same is true for MFP given a large aperture VLA.) However, as remarked above, to estimate the source depth, one must estimate the source range first or search for range and depth simultaneously, which requires knowing the mode wavenumbers to a high precision (third decimal point). As the mode wavenumbers are very sensitive to the environmental acoustic inputs, when the calculated mode wavenumbers k'_m does not match the true wavenumbers k_m , the MFP/MMP depth ambiguity function becomes $\text{DMMP}(r = r_s, z) = \left| \sum_{m=1}^M \phi_m(z) \phi_m(z_s) e^{i(k'_m - k_m)r_s - 2\alpha_m r_s} / k_m \right|^2$. In this case, the depth ambiguity function is still a function of the mode wavenumbers, and has the environmental mismatch problem. Likewise, when the assumed source range is different from the true source range, the MMP/MFP depth ambiguity function is a function of the mode wavenumbers (with or without environmental mismatch) and is expected to yield a wrong source depth. In comparison, the proposed method, Eq. (9), does not require knowledge of the source range and is not an explicit function of the mode wavenumbers.

C. Requirements and resolution

The above analyses are based on Eq. (1), which is often referred to as the conventional (synthetic range) beamforming. Conventional (synthetic range) beamforming usually requires a large range span; to resolve the wavenumbers between the n and m th modes, the range needs to be larger than the mode interference distance $\lambda_{nm} = 2\pi/(k_n - k_m)$. One finds, for example, $\lambda_{12} \sim 2.3$ km; higher order modes requires a smaller mode interference distance to resolve them. Practical problems may present data with a short range span, e.g., 1 km, for which one must apply the high resolution algorithms,¹² which are capable of resolving modes with a less range span but may have the snapshot deficient problem. Data with a short range span can often be considered range-independent, presenting an additional incentive for the high resolution algorithms.

Similarly, Eqs. (6), (9), or (10) belongs to a class of the Bartlett processor. The depth resolution is in this case determined by the mode vertical wavenumbers, and has been widely studied in the literature for various environments, including that using MFP/MMP. To improve the depth resolution, one can also apply a high resolution method.¹³ These topics will be pursued in future work.

V. SUMMARY AND DISCUSSIONS

A method for estimating the source depth is presented for a single receiver based on the synthetic aperture created by a moving source relative to the receiver. The received signal as a function of range is steered by a range-dependent phase and beamformed. The beamformed output yields the modal wavenumber spectrum; the spectral peaks contain information about the source depth. A source–depth ambiguity function is proposed by multiplying the modal spectral peaks with the mode depth functions, given proper identification of the modes. Note that the mode depth function is insensitive to SSP variations. Mode depth functions calculated using the nominal SSP is found good enough for coarse estimation of the source depth.

This method is tested with simulated data, and real data with reasonable success; for the latter, we choose the SWellex96 data with a sufficiently high SNR. For real data, the source jittering motion (due to ocean waves) is known to introduce instantaneous acceleration and deceleration, injecting a random phase to the signal. This creates Doppler spread for a CW signal, and severely impacts the performance of synthetic beamforming. In this paper, a PLL is used to remove the random phase components of the signal due to source jittering motion and random media fluctuations. Modal wavenumber spectrum is reasonably recovered from the data.

Given a VLA covering the (upper) water column including the depths where the source is expected to travel, the beam power from individual hydrophones yields the depth ambiguity function as a function of depth. No information on the environment, such as SSP or bottom profile is needed. In other words, it is free of the environmental mismatch problem.

As remarked in the Introduction, to estimate the source depth using MMP, or MFP, one needs to estimate the source range and depth simultaneously, else one must know the source range to determine the source depth and vice versa. The advantage of the proposed method, Eqs. (6) or (9), is that the source range is not known nor is it a required input. The disadvantage of this method is that it requires a moving source traveling a sufficient radial distance whereas MMP/MFP can be applied to a fixed as well as a moving source. More effort is needed for application to real world problems as remarked in Sec. IV, which is left for future work.

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APPENDIX: ESTIMATING THE NARROWBAND SPECTRAL DENSITY FOR A MOVING SOURCE

The time series for a narrowband signal can be expressed as follows:

$$p(r, z_0, t) = b(t) \sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r}} \phi_m(z_s) \phi_m(z_0) \times e^{-i(2\pi f_c t - k_m r) - \alpha_m r - i\pi/4} + n(t), \quad (\text{A1})$$

where f_c is the signal frequency, $b(t)$ is a random variable reflecting the effect of the random medium and the source motion, and $n(t)$ denotes the ambient noise. Dividing the data into blocks, each with a duration T and containing L samples, i.e., $T = L\Delta t$, where Δt is the sample duration, one can express time for the l th sample in the n th block as $t_{n,l} = T_n + l\Delta t$, where $T_n \equiv nT$, $l = 1, 2, \dots, L$. For a moving source with a radial velocity of v , the range can be expressed as $r_{n,l} = r_n + l\Delta r$, where $r_n = vT_n$ and $\Delta r = v\Delta t$. The time series for a moving source can be expressed as¹⁴

$$p(r, z_0, t_{n,l}) \simeq e^{i\hat{\theta}(t_{n,l})} \sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r_n}} \phi_m(z_s) \phi_m(z_0) \times e^{-i(2\pi f_c T_n - k_m r_n) - \alpha_m r_n - i\pi/4} e^{-i2\pi f_D(l\Delta t)} + n(t_{n,l}), \quad (\text{A2})$$

where $f_D = f_c(1 + v/c)$ is the Doppler shifted frequency. In deriving Eq. (A2), we have: (1) assumed a constant Doppler shift for the dominant modes, namely, $k_m \Delta r \simeq 2\pi f_c (v/c) \Delta t$, where $c = 2\pi f_c / \bar{k}_m$, $\bar{k}_m$ is wavenumber averaged over the dominant modes; (2) expressed $b(t) = |b|e^{i\hat{\theta}(t)}$, and assumed $|b| \simeq 1$, since the amplitude varies much more slowly with phase, and has a small effect on the spectral estimation compared with the phase; and (3) approximated $e^{-\alpha_m l\Delta r} \simeq 1$, since $l < L$ and $\alpha_m L\Delta r \ll 1$. As stated in the text, we assume that the random phase $\theta(t)$ is caused by source jittering motion and random medium and can be filtered out by the PLL.

Applying PLL to estimate the random phase $\hat{\theta}(t)$ (see Sec. III) and carrying out the Fourier transform for the n th block of data (for a given r_n), one obtains an estimate of the pressure field,

$$\begin{aligned} \tilde{p}(r_n, z_0, f) &= \sum_{l=1}^L e^{i2\pi f t_{n,l}} \left(e^{-i\hat{\theta}(t_{n,l})} p(r_{n,l}, t_{n,l}) \right) \\ &= e^{i2\pi(f-f_c)T_n} \sum_{l=1}^L e^{i2\pi f(l\Delta t)} e^{-i2\pi f_D(l\Delta t)} e^{i\Delta\theta(t_{n,l})} \\ &\quad \times \sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r}} \phi_m(z_s) \phi_m(z_0) e^{ik_m r_n - \alpha_m r_n - i\pi/4}, \end{aligned} \quad (\text{A3})$$

where $\hat{\theta}(t_{n,l})$ is the estimated phase, and $\Delta\theta(t_{n,l}) = \theta(t_{n,l}) - \hat{\theta}(t_{n,l})$ is the phase estimation error. One then finds for $f = f_0$,

$$\tilde{p}(r_n, z_0, f_D) = L e^{i2\pi(\Delta f)T_n} e^{i\bar{\Delta}\theta(T_n)} p(r_n, z_0), \quad (\text{A4})$$

where $\Delta f = f_D - f_c$ is the Doppler frequency shift, and $p(r_n, z_0)$ is pressure field given in Eq. (3), and we have approximated the phase error, assuming it is small, using $\sum_{l=1}^L e^{i\Delta\theta(t_{n,l})} \simeq L e^{i\bar{\Delta}\theta(T_n)}$, where $\bar{\Delta}\theta(T_n) = (1/L) \sum_{l=1}^L \Delta\theta(t_{n,l})$. Equation (A4) states that the pressure field $p(r_n, z_0)$ can be estimated from the moving source data, block by block, by correcting for the Doppler phase for each block using $e^{-i2\pi(\Delta f)T_n} \tilde{p}(r_n, z_0, f_D)$, if there is no error in the random phase estimation, $\Delta\theta = 0$. The pressure field $p(r_n, z_0)$ can then be used to estimate the mode wavenumber spectrum using Eq. (1) as demonstrated in Sec. III.

One finds that the mode wavenumber spectrum in real data processing is sensitive to small errors in the estimation of the Doppler shift, since a small change/error in Δf , e.g., $\delta(\Delta f) = 0.05$ Hz, produces a large phase shift $\delta(\Delta f)T_n$ over a long time period. While the shape (or the mode structure) of the wavenumber spectrum remains unchanged for different Δf , or different sizes of Fourier transform, the mode wavenumbers (the *position* of the spectrum) are sensitive to small changes in Δf . This is because a linear phase shift with respect to range (time) produces a delay shift in the wavenumber. In this regard, one notes that the slope of the filtered phase in Fig. 3(c) or Fig. 3(d) is about 1.1 times smaller than that expected from the estimated Doppler shift—the difference is partially due to unwrapping errors [see phase jumps in Fig. 3(c)] as well as uncertainty in the Doppler estimation. Second, in the above derivation we have assumed a constant Doppler shift for all the (dominant) modes, but in reality, different modes have slightly different Doppler shifts, producing effectively a (small) Doppler spread. The Doppler shift estimated from the spectrum has an uncertainty on the order of the maximum of the Doppler spread and the inverse of the Fourier transform window. For the present case Δf is known to the first decimal point but not precisely known to the second decimal point.

Equation (6) was used to estimate the source depth. Note that Eq. (6) uses only the spectrum amplitudes not the corresponding mode wavenumbers. To use Eq. (6), one needs to associate the spectrum peaks with the proper modes. To do so, one can adjust the Doppler shift until the measured spectrum grossly lines up with the modeled spectrum—as remarked above the spectrum itself (the structure of the modes and the mode amplitudes) is insensitive to the Doppler shift error. While there is an uncertainty in estimating the mode wavenumbers it will not affect the depth estimation so long as modes are properly identified (using the simulated wavenumber spectrum as a guide).

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